

**A PROJECT TITLE**  
**ON**  
**COVID-19 Detection Using CNN and VGG16 on Chest X-Rays**

*project report submitted in partial fulfilment of the requirements for the Course*

**MACHINE LEARNING**  
**BACHELOR OF TECHNOLOGY**  
**in**  
**A.Y.: 2026-27**

**by**

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**CSE**



**Koneru Lakshmaiah Education Foundation**

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

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## DEPARTMENT OF CSE

Name of Title: **COVID-19 Detection Using CNN and VGG16 on Chest X-Rays**

### 1. Abstract:

The COVID-19 pandemic affected millions of people around the world and created a need for fast and reliable methods for detecting the disease. Early detection is important because it helps patients receive proper treatment at the right time and also helps in controlling the spread of the infection. Chest X-ray is one of the commonly used medical imaging techniques that can provide information about the condition of a patient's lungs. With the development of deep learning, it is possible to use computer-based models to study chest X-ray images and identify patterns related to COVID-19. In this project, we are developing a COVID-19 detection system using Convolutional Neural Network (CNN) and VGG16. The COVID-19 Radiography Database from Kaggle will be used as the dataset for training and testing the model. It contains chest X-ray images of COVID-19, Normal, Viral Pneumonia, and Lung Opacity cases. The images will first be pre-processed by resizing and normalizing them, and data augmentation will be applied to increase the number and variety of training images. CNN will learn important features from the X-ray images, while VGG16, which is a pre-trained deep learning model, will be used to improve the classification process. The trained model will then classify the chest X-ray images and identify whether they belong to the COVID-19 or non-COVID-19 category. The performance of the model will be evaluated using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC Score. The main aim of this project is to develop an easy-to-use and effective deep learning-based system that can help in the early screening of COVID-19 and provide useful support to healthcare professionals.

### 2. INTRODUCTION:

The motivation for this project comes from the need for faster and more reliable methods of detecting COVID-19, especially when medical resources and diagnostic facilities are limited. During the COVID-19 pandemic, early identification of infected patients became important for timely treatment and reducing the spread of infection. Chest X-ray images are widely used in medical diagnosis because they are relatively accessible and can provide useful information about lung conditions. Therefore, developing an automated system that can assist in identifying COVID-19 from chest X-ray images can support healthcare professionals in the screening process.

This project focuses on COVID-19 detection from chest X-ray images using Convolutional Neural Networks (CNN) and the VGG16 deep learning model. CNNs are well suited for image classification because they can automatically learn important visual features such as patterns, shapes, and abnormalities from images. VGG16 is a pretrained deep learning architecture that can be used through transfer learning to classify chest X-ray images into relevant categories. The proposed system aims to train and evaluate these models on a suitable chest X-ray dataset and compare their performance using measures such as accuracy, precision, recall, and F1-score. The project demonstrates how deep learning and computer vision can be applied to assist medical image analysis and provide a computer-aided approach to COVID-19 screening.

### 3. SOFTWARE REQUIREMENTS:

| Category                | Requirement / Description                                                                                                   |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Operating System        | Windows 10/11, Linux, or any compatible operating system                                                                    |
| Programming Language    | Python 3.x for implementing deep learning models, image preprocessing, training, and evaluation                             |
| Development Environment | Jupyter Notebook, Google Colab, or Visual Studio Code                                                                       |
| Deep Learning Framework | TensorFlow and Keras for building, training, and evaluating CNN and VGG16 models                                            |
| NumPy                   | Used for numerical computations and array operations                                                                        |
| Pandas                  | Used for handling and organizing dataset information                                                                        |
| OpenCV                  | Used for image processing and preprocessing                                                                                 |
| Matplotlib              | Used for visualizing images, training performance, and results                                                              |
| Scikit-learn            | Used for Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC Score                                         |
| Dataset                 | COVID-19 Radiography Database from Kaggle containing COVID-19, Normal, Viral Pneumonia, and Lung Opacity chest X-ray images |
| Pre-trained Model       | VGG16 pre-trained on ImageNet for extracting image features and classification                                              |
| Data Storage            | Local computer storage or Google Drive for storing datasets, trained models, and project files                              |
| Web Browser             | Google Chrome, Microsoft Edge, or another modern browser for accessing Google Colab, Kaggle, and project resources          |

Signature of the Guide

Course Instructor